Corporate Finance Formula Sheet (W1 to W4)

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Every formula is tagged W1, W2, W3 or W4 for the week of your PDFs (CFW1 to CFW4), followed by solved examples with steps.

Formula sheet: W1 (assets, markets, 2008 crisis, structured finance)

Tag	Concept	Formula	Notes
W1	One-year bond return	i = (Face - P) / P	Higher price means lower return
W1	One-year bond price	P = Face / (1 + i)	Price falls when rates rise
W1	Stock price from next-year value	P = (Value next year) / (1 + required return)	Higher systematic risk means higher required return, so lower price
W1	Earnings per share	EPS = Profit / Number of shares	Your share = EPS x shares you own
W1	Dividend and retained earnings	Dividend = payout % x your profit share; Retained = profit share - dividend	Both belong to shareholders
W1	P/E ratio	P/E = Price / EPS	Shiller's bubble signal
W1	Order book spread	Spread = best ask - best bid	Buy fills at ask, sell fills at bid
W1	Annual growth over n years	g = (End / Start)^(1/n) - 1	Used for the Nasdaq 750 to 5,000 case
W1	Percentage fall from peak	Fall = 1 - Bottom / Peak	
W1	Leverage	L = Total assets / Equity	
W1	Asset fall that wipes out equity	1 / L	At L = 20 it is 5%

Tag	Concept	Formula	Notes
W1	Expected loss	EL = P(default) x Loss given default	LGD = 1 - recovery rate
W1	Two independent bonds, same p	Junior default = 1 - (1-p)^2; Senior default = p^2	Junior fails if either defaults
W1	n independent bonds	Junior = 1 - (1-p)^n; Senior = p^n	Three bonds, middle = 3p^2(1-p) + p^3
W1	Two bonds with default correlation rho	P(both default) = p^2 + rho x p(1-p)	rho = 0 gives p^2; rho = 1 gives p
W1	CDO squared senior tranche	P = (Junior default prob)^2	Two independent junior tranches pooled
W1	Rating grades	AAA, AA, A, BBB = investment grade; BB and below = speculative	

Formula sheet: W2 (returns, time value, NPV, annuities, bonds)

Tag	Concept	Formula	Notes
W2	Rate of return	r = (C1 + Div - C0) / C0	Div = interim payment
W2	Return split	r = Capital gain (C1 - C0)/C0 + Dividend yield Div/C0	Can be positive even if price falls
W2	Basis points	1 bp = 0.01%; 100 bp = 1 percentage point	5% to 10% = 500 bp
W2	Future value	FV = C0 x (1 + r)^t	Compounding = interest on interest
W2	t-year holding return	1 + r_t = (1 + r)^t	
W2	Rate from a multi-year return	r = (1 + total return)^(1/t) - 1	50% over 2 years = 22.47% a year
W2	Years to double	x = ln 2 / ln(1 + r)	Rule of 72: years ~ 72 / (rate in %)
W2	Sub-period rate	(1 + r_half)^2 = 1 + r	Half-year rate at 5% a year = 2.47%

Tag	Concept	Formula	Notes
W2	Present value	PV = Ct / (1 + r)^t	Discount factor = 1/(1+r)^t
W2	NPV	NPV = C0 + C1/(1+r) + C2/(1+r)^2 + ...	C0 negative; accept if NPV > 0
W2	Intertemporal budget line	c1 + c2/(1+r) = m1 + m2/(1+r)	Slope = -(1+r)
W2	Max c1 and max c2	Max c1 = m1 + m2/(1+r); Max c2 = m2 + (1+r) m1	
W2	Real interest rate	1 + rho = (1 + r) / (1 + pi)	Approx r - pi only if small
W2	Budget line with inflation	c2 = -[(1+r)/(1+pi)] c1 + [m1(1+r) + m2]/(1+pi)	Corrected version from CF_2A_v2
W2	No-arbitrage price	p0 = p1 / (1 + r)	Price = present value
W2	Continuous compounding	FV = C e^(rT); PV = C e^(-rT)	Limit of (1 + r/n)^(nT)
W2	Forest cutting rule	Cut when V′(T)/V(T) = r	Growth rate = interest rate
W2	Perpetuity	PV = C / r	First payment at t = 1; if first payment today, PV = C + C/r
W2	Growing perpetuity	PV = C1 / (r - g)	Need r > g
W2	Annuity	PV = (C/r) [1 - 1/(1+r)^T]	First payment at t = 1
W2	Zero-coupon bond	P = Face / (1 + r)^T	
W2	Coupon bond	P = annuity of coupons + Face/(1+r)^T	Coupon rate < market rate means discount; > means premium

Formula sheet: W3 (loans, Gordon model, IRR, PI, payback, varying rates)

Tag	Concept	Formula	Notes
W3	Loan / annuity payment	Loan = (C/r) [1 - 1/(1+r)^n]	r and n in the same period (months)
W3	Quoted annual rate	Monthly rate = quoted rate / 12	No compounding in the quote

Tag	Concept	Formula	Notes
W3	Interest in a payment	Interest = rate x outstanding balance	Principal = payment - interest
W3	Constant perpetuity	PV = C / r	Rough business value
W3	Gordon growth model	P = D / (r - g)	D = next year's dividend
W3	Infer cost of capital	D/P = r - g, so r = D/P + g	Dividend yield plus growth
W3	NPV	NPV = Sum Ct / (1+r)^t	Take if > 0
W3	IRR	Rate where NPV = 0	Take if IRR > cost of capital; unreliable with sign changes or mutually exclusive projects
W3	Profitability Index	PI = PV(future cash flows) / Cost	PI > 1 same as NPV > 0; biased to small projects
W3	Payback period	Years until initial cost recovered	Ignores later flows and time value
W3	Two-period holding return	1 + r0,2 = (1 + r0,1)(1 + r1,2)	Multiply, do not add
W3	Annualized return	1 + annualized = (1 + holding return)^(1/t)	t-th root
W3	NPV with varying rates	NPV = C0 + C1/(1+r0,1) + C2/(1+r0,2) + ...	Each flow uses its own rate

Formula sheet: W4 (duration, inflation, yield curve)

Tag	Concept	Formula	Notes
W4	Duration	D = Sum(Ct x t) / Sum(Ct)	Balance point of payments
W4	Macaulay duration	D = Sum(PV(Ct) x t) / Sum(PV(Ct))	Uses present values; depends on rates
W4	CPI	CPI = (Basket cost now / Basket cost in base year) x 100	Base-year quantities fixed

Tag	Concept	Formula	Notes
W4	CPI as weights	CPI/100 = Sum f_i x (p_i now / p_i base)	f_i = base-year spending share
W4	Nominal GDP	Sum of current price x current quantity	
W4	Real GDP	Sum of base price x current quantity	Changes only with quantities
W4	GDP deflator	Nominal GDP / Real GDP	2.0 means prices doubled
W4	Fisher relation	1 + r = (1 + i) / (1 + expected inflation)	Approx r = i - inflation; use exact if rates large
W4	NPV with inflation	Nominal flows with nominal rate; real flows with real rate	Never mix
W4	Holding vs annualized	1 + r0,t = (1 + r_t)^t	Yield curve shows annualized r_t
W4	Interpolation	Missing yield ~ average of neighbouring yields	Approximate
W4	Forward rate	1 + r1,2 = (1 + r2)^2 / (1 + r1)	Three years: (1+r3)^3 = (1+r1)(1+r1,2)(1+r2,3)
W4	Zero bond price	P = Face / (1 + r_t)^t	Long maturity means price more sensitive to rate
W4	Price after one year	Price = Face / [(1 + forward rates for the remaining years)]	One-year return equals the one-year rate under certainty
W4	Term premium	Long-term rate - short-term rate	Compensation for risk
W4	Extra from long bond vs rolling	(1+r2)^2 / [(1+r1)(1 + expected next 1-year rate)] - 1	Term premium over two years

Solved examples: W1

W1 Ex 1: Bond return and price

Formulas (W1): i = (Face - P)/P and P = Face/(1 + i). A bond pays Rs. 10,000 in a year and costs Rs. 9,500. Find the yield, and the fair price of a Rs. 5,000 bond when the market rate is 12%.

1. Profit = 10,000 - 9,500 = 500.
2. i = 500 / 9,500 = 5.26%.
3. Price at 12%: P = 5,000 / 1.12 = Rs. 4,464.29.

W1 Ex 2: Leverage

Formulas (W1): L = Assets / Equity; wipe-out fall = 1/L. A bank has assets 100 and equity 4.

1. L = 100 / 4 = 25.
2. Fall that wipes out equity = 1/25 = 4%.
3. If assets fall 3%, loss = 3, so equity = 4 - 3 = 1, which is 25% of the original equity.

W1 Ex 3: Tranches with independent bonds

Formulas (W1): Junior = 1 - (1-p)^n; Senior = p^n; Middle (3 bonds) = 3p^2(1-p) + p^3. Each bond defaults with p = 20%.

1. Two bonds: Junior = 1 - 0.8^2 = 1 - 0.64 = 36%. Senior = 0.2^2 = 4%.
2. Three bonds: Junior = 1 - 0.8^3 = 48.8%.
3. Middle = 3(0.04)(0.8) + 0.008 = 0.096 + 0.008 = 10.4%.
4. Senior = 0.2^3 = 0.8%. Middle and senior are both safer than a single 20% bond.

W1 Ex 4: Correlation and CDO squared

Formulas (W1): P(both) = p^2 + rho x p(1-p); CDO squared senior = (junior default)^2. Two bonds with p = 10%, rho = 0.2.

1. P(both default) = 0.01 + 0.2 x (0.1 x 0.9) = 0.01 + 0.018 = 2.8%.
2. Compare: rho = 0 gives 1%; rho = 1 gives 10%.
3. CDO squared from two independent junior tranches of 36% each: 0.36^2 = 12.96%.

W1 Ex 5: Order book and expected loss

Formulas (W1): buy fills at the ask; EL = P(default) x LGD. Asks: 100 at 101.00, 200 at 101.10, 300 at 101.50. You place a limit buy for 250 shares at 101.10.

1. Fill the 100 shares at 101.00 = 10,100.
2. Fill the remaining 150 at 101.10 = 15,165.
3. Total cost = Rs. 25,265 (average 101.06).
4. Expected loss for a bond with a 4% default chance and 60% recovery: LGD = 40%, so EL = 4% x 40% = 1.6%.

Solved examples: W2

W2 Ex 1: Return with a dividend

Formula (W2): r = (C1 + Div - C0)/C0 = capital gain + dividend yield. A share costs Rs. 92, pays Rs. 5, then is worth Rs. 110.

1. Total return = (110 + 5 - 92)/92 = 23/92 = 25%.
2. Capital gain = 18/92 = 19.6%.
3. Dividend yield = 5/92 = 5.4%. The two parts add to 25%.

W2 Ex 2: Compounding and reverse questions

Formulas (W2): FV = C0(1+r)^t; r = (1 + total)^(1/t) - 1.

1. Rs. 1,000 at 8% for 5 years: 1,000 x 1.08^5 = Rs. 1,469.33.
2. A 2-year return of 44% needs (1+r)^2 = 1.44, so r = 20% a year.
3. Six-month rate for a 6% annual rate: sqrt(1.06) - 1 = 2.96%, not 3%.
4. Years to double at 7%: ln 2 / ln 1.07 = 10.24 (Rule of 72 gives 10.3).

W2 Ex 3: NPV

Formula (W2): NPV = C0 + Sum Ct/(1+r)^t. A project costs 500 and pays 200, 250, 300 at the ends of years 1 to 3; r = 10%.

1. PV year 1 = 200/1.10 = 181.82.
2. PV year 2 = 250/1.21 = 206.61.
3. PV year 3 = 300/1.331 = 225.39.
4. NPV = 181.82 + 206.61 + 225.39 - 500 = +113.82, so accept.
5. At r = 25% the PVs are 160 + 160 + 153.60 = 473.60, NPV = -26.40, so reject.

W2 Ex 4: Perpetuity, growing perpetuity, annuity

Formulas (W2): C/r; C1/(r-g); (C/r)[1 - 1/(1+r)^T].

1. Rs. 60 a year forever at 8%: 60/0.08 = Rs. 750.

2. Rs. 60 next year growing 3% at 8%: 60/(0.08 - 0.03) = Rs. 1,200.
3. Rs. 1,000 a year for 5 years at 10%: (1,000/0.10) x [1 - 1/1.1^5] = 10,000 x 0.37908 = Rs. 3,790.79.

W2 Ex 5: Bond pricing

Formula (W2): P = annuity of coupons + Face/(1+r)^T. A 3-year bond pays a Rs. 50 coupon yearly and Rs. 1,000 principal; r = 5%.

1. 50/1.05 = 47.62; 50/1.05^2 = 45.35; 1,050/1.05^3 = 907.03.
2. Price = 47.62 + 45.35 + 907.03 = Rs. 1,000.
3. Coupon rate equals market rate, so the bond trades at par.

W2 Ex 6: Real rate, budget line, no-arbitrage

Formulas (W2): 1 + rho = (1+r)/(1+pi); c2 = m2 + (1+r)(m1 - c1); p0 = p1/(1+r).

1. Nominal 10%, inflation 5%: 1 + rho = 1.10/1.05 = 1.0476, so rho = 4.76%.
2. With m1 = 100, m2 = 110, r = 10%: max c1 = 100 + 110/1.1 = 200; max c2 = 110 + 1.1 x 100 = 220.
3. If c1 = 150: c2 = 110 + 1.1 x (100 - 150) = 55.
4. Asset certain to be worth 110 next year, bank rate 5%: fair price = 110/1.05 = Rs. 104.76.

Solved examples: W3

W3 Ex 1: Loan payment

Formulas (W3): Loan = (C/r)[1 - 1/(1+r)^n]; Interest = rate x balance. Borrow 200,000 for 20 years at a quoted 9%, monthly payments.

1. Monthly rate = 9% / 12 = 0.75%; n = 20 x 12 = 240.
2. Payment C = 200,000 x 0.0075 / [1 - 1.0075^(-240)] = 1,799.45.
3. Month 1 interest = 0.75% x 200,000 = 1,500.00.
4. Principal repaid = 1,799.45 - 1,500.00 = 299.45.

W3 Ex 2: Gordon growth model

Formulas (W3): P = D/(r - g); r = D/P + g. A stock pays 50 next year, growing 4% forever, required return 9%.

1. P = 50 / (0.09 - 0.04) = 50 / 0.05 = 1,000.
2. Reverse: if D = 10, P = 200, g = 5%, then r = 10/200 + 5% = 10%.

W3 Ex 3: NPV, PI, payback and IRR

Formulas (W3): NPV = Sum Ct/(1+r)^t; PI = PV(future flows)/Cost; IRR solves NPV = 0. Cost of capital 10%. Project A: -100, +60, +60. Project B: -100, +10, +20, +200.

1. A: PV = 60/1.1 + 60/1.21 = 54.55 + 49.59 = 104.13; NPV = 4.13; PI = 1.04; payback = 2 years.
2. B: PV = 9.09 + 16.53 + 150.26 = 175.88; NPV = 75.88; PI = 1.76; payback = 3 years.
3. Choose B: much higher NPV even though A pays back sooner.
4. IRR of -500, +200, +250, +300 is about 21.65% (trial and error or Excel). It exceeds 10%, so accept.
5. IRR trap: invest 5 get 10 (IRR 100%, NPV 4.52 at 5%) vs invest 1,000 get 1,100 (IRR 10%, NPV 47.62). Pick the higher NPV.

W3 Ex 4: Holding return and annualized return

Formulas (W3): 1 + r0,2 = (1 + r0,1)(1 + r1,2); annualized = (1 + holding)^(1/t) - 1. Yearly returns 10%, 20%, 5%.

1. Holding: 1.10 x 1.20 x 1.05 = 1.386, so 38.6%.
2. Annualized: 1.386^(1/3) - 1 = 11.49% (not 38.6/3 = 12.87%).
3. Averages mislead: +50% then -50% averages 0%, but 100 becomes 75. Annualized = sqrt(0.75) - 1 = -13.40% a year.

W3 Ex 5: NPV with different rates

Formula (W3): NPV = C0 + C1/(1+r0,1) + C2/(1+r0,3). Cost 50 now; pays 30 in 1 year and 40 in 3 years. The 1-year rate is 4% and the 3-year annualized rate is 5%.

1. PV of 30 = 30 / 1.04 = 28.85.
2. PV of 40 = 40 / 1.05^3 = 34.55.
3. NPV = -50 + 28.85 + 34.55 = +13.40, so accept.

Solved examples: W4

W4 Ex 1: Duration

Formulas (W4): D = Sum(Ct x t)/Sum(Ct); Macaulay = Sum(PV x t)/Sum(PV).

1. 50 in year 1 and 50 in year 10: D = (50x1 + 50x10)/100 = 5.5 years.
2. 100 in year 1 and 100 in year 2, spot rates 4% (1y) and 5% (2y): PV1 = 100/1.04 = 96.15; PV2 = 100/1.05^2 = 90.70.
3. Macaulay = (96.15 x 1 + 90.70 x 2) / 186.85 = 1.485 years.

W4 Ex 2: CPI, GDP and deflator

Formulas (W4): CPI = basket now / basket base x 100; Real GDP = base prices x current quantities; Deflator = Nominal / Real.

1. Basket 20 rice, 10 milk, 5 fuel; base prices 50, 40, 80; new prices 60, 44, 96. Base cost = 1,000 + 400 + 400 = 1,800; current cost = 1,200 + 440 + 480 = 2,120.
2. CPI = 2,120 / 1,800 x 100 = 117.78, so inflation is 17.78%. Rice weight = 1,000/1,800 = 55.6%.
3. Laptop 20, car 50 (base prices). Last year 30 laptops and 10 cars; this year prices 25 and 60, output 40 and 12. Nominal last year = 1,100; nominal this year = 40x25 + 12x60 = 1,720.
4. Real this year = 40x20 + 12x50 = 1,400. Deflator = 1,720/1,400 = 1.229. Real growth = 1,400/1,100 - 1 = 27.3%.

W4 Ex 3: Real rate and inflation inside NPV

Formulas (W4): 1 + r = (1+i)/(1+inflation); nominal flows with nominal rate, real flows with real rate.

1. Nominal 12%, expected inflation 5%: r = 1.12/1.05 - 1 = 6.67% (approx 7%).
2. Large rates: nominal 300%, inflation 100%: r = 4/2 - 1 = 100%, not 200%.
3. Project pays 1,100 next year, nominal rate 10%, inflation 4%. Nominal way: 1,100/1.10 = 1,000.
4. Real way: real flow = 1,100/1.04 = 1,057.69; real rate = 1.10/1.04 - 1 = 5.77%; PV = 1,057.69/1.0577 = 1,000. Same answer. Mixing (1,100 at 5.77%) gives about 1,040, which is wrong.

W4 Ex 4: Yield curve, forward rates and bond price risk

Formulas (W4): 1 + r0,t = (1+r_t)^t; 1 + r1,2 = (1+r2)^2/(1+r1); P = Face/(1+r)^t.

1. 3-year yield 0.97%: 50,000 x 1.0097^3 = 51,469.16; holding return = 2.94%.
2. Interpolate 15-year yield from 10y (1.52%) and 20y (1.94%): (1.52 + 1.94)/2 = 1.73%.
3. 1y = 3%, 2y = 4%, 3y = 5%. Forward r1,2 = 1.04^2/1.03 - 1 = 5.01%. Forward r2,3 = 1.05^3/1.04^2 - 1 = 7.03%.
4. Two-year zero paying 100: price today = 100/1.04^2 = 92.46; after one year = 100/1.0501 = 95.23; return = 3.00%, equal to the 1-year rate.
5. 20-year zero paying 1,000 at 2%: price = 1,000/1.02^20 = 672.97. At 2.1% it is 659.91 (-1.94%); at 1.9% it is 686.30 (+1.98%). A 1-year zero moves only about 0.1%.
6. Term premium: 1-year 5%, 2-year 10%, expected next 1-year rate 8%. Rolling = 1.05 x 1.08 = 1.134; 2-year bond = 1.10^2 = 1.21; extra = 1.21/1.134 - 1 = 6.7%.